Predicting Full-Benchmark LLM Scores from a Few Items: A Controlled Comparison and a Model-Assisted Sampling Estimator

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Abstract

Predicting a language model's full-benchmark score from its results on a few items, using per-item results of other models, could cut evaluation cost, but published methods were compared under different protocols. We evaluate 13 binary-score benchmarks (Open LLM Leaderboard v2, HELM) with one protocol: equal budgets (10–200 items), source/target splits that are random, extrapolating (the strongest 20% of models as targets) or whole-organisation, source-only selection and tuning, duplicate models removed, and model-level bootstrap intervals with paired comparisons. We re-implement existing and simple methods and add MASS, a model-assisted sampling estimator (stratified probability-proportional-to-size item design, latent-trait predictor, design-weighted correction on the sampled items). Against the random-subset mean at 50–100 items, MASS-Ens lowers RMSE by roughly 21–27% (all 13 benchmarks; effective-sample-size gain about 1.6–1.8). It is not distinguishable from the best baseline chosen by cross-validation on source models, nor from ridge regression on a random subset when targets resemble the sources, and it does not improve the ranking of target models. Under extrapolation, predictors fitted on weaker models degrade (ridge: RMSE 1.4–1.7 times the random mean), and the difference correction is what prevents this. Its reported intervals under-cover at 10–20 items. The evidence supports "effective under conditions": useful for estimating scores, not for ranking the strongest models.

1 Introduction

Evaluating a language model on a benchmark means running it on every item, and the cost multiplies when many checkpoints, benchmarks, prompts or inference settings are compared. A natural shortcut is to run the model on a small subset and predict its full-benchmark score. Historical results of other (*source*) models on every item are usually available and could guide which items to keep and how to extrapolate from them. Several methods build on this idea: clustering items into anchors [Vivek et al., 2024], IRT-based prediction [Maia Polo et al., 2024a, Kipnis et al., 2025], learned predictors and selection policies [Pacchiardi et al., 2024, Li et al., 2025]. They were however evaluated on different model pools, splits, metrics and numbers of repetitions, often with few repetitions.

We ask how much the source models' results actually help, and when. Using the 0/1 item-level results of 13 benchmarks (Open LLM Leaderboard v2 and HELM), we run one protocol for all methods: the same budgets, the same source/target splits, selection from source models only, exact duplicate models removed, and uncertainty from model-level bootstrap with paired comparisons. Besides random splits of models (interpolation) we hold out the *strongest* 20% of models as targets (extrapolation)

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and, for the Open LLM benchmarks, whole organisations. We re-implement the methods ourselves, and add a model-assisted sampling estimator, MASS, that adapts the difference estimator of survey sampling [Särndal et al., 1992] to the benchmark setting: a stratified, probability-proportional-to-size item design built from source models, a latent-trait predictor, and a design-weighted correction on the sampled items.

Contributions and what we find. (1) A common, stricter protocol and re-implementations of the published methods we could reproduce and of simple baselines (Appendix A); the data-repository conclusions of Zhang et al. [2025] are tested as hypotheses and are confirmed or qualified in Section 4.2. (2) MASS-Ens, whose design is a straightforward combination of known survey-sampling and prediction-powered ideas [Särndal et al., 1992, Angelopoulos et al., 2023]; its value is empirical. With 50–100 target items it lowers the RMSE of the full-score prediction by about 21–27% relative to the random-subset mean under extrapolation and interpolation, on all 13 benchmarks, an effective-sample-size gain of about 1.6–1.8. (3) Limits of that result: it is not distinguishable from the best baseline chosen by cross-validation on source models, nor from ridge regression on a random subset under interpolation, where ridge has the better point estimate; the lower error does not improve the ranking of target models; under extrapolation it is a small effect compared with the spread of the strongest models; and its reported intervals under-cover at small budgets. (4) A decomposition showing that under extrapolation the difference correction on the sampled items is what prevents the degradation of predictors fitted on weaker models, with design and predictor each adding a little. We conclude that subset-based prediction is effective under conditions: useful for estimating a score to a few accuracy points when the new model resembles the source models or a correction term is used, and not a reliable tool for ranking the strongest models.

2 Related Work

Selecting items from historical results. Anchor Points [Vivek et al., 2024] clusters items by how a pool of source models responds to them, keeps the cluster medoids as anchors and estimates a target's score as a cluster-size-weighted average over the anchors. tinyBenchmarks [Maia Polo et al., 2024a] combines such anchors with item-response-theory (IRT) models fitted on the source models, and releases reduced benchmarks; metabench [Kipnis et al., 2025] selects items by IRT information and predicts with a generalised additive model; Pacchiardi et al. [2024] train an "assessor" on the target's results over a reference set, and Li et al. [2025] learn an adaptive item-acquisition policy; Balkır et al. [2026] adapt IRT-based adaptive testing to continuous scores and ranking. These works use different model pools, splits (random, by date, by organisation), metrics and numbers of repetitions, so their reported errors are not comparable with each other; our contribution on this axis is a single protocol over the same matrices, with the same budget, source/target split and metrics for every method.

The closest comparison. Zhang et al. [2025] compare eleven benchmark-prediction methods over many benchmarks and release the score matrices and implementations that we use (Section 4.1). Their repository states conclusions about which methods are competitive and which fail when targets are stronger than the source models; we treat those statements as hypotheses to be tested, not as inputs. We re-implement all methods ourselves (the released `benchpred` package is inspected but not used), remove exact duplicate model rows before splitting, add a family-held-out split, report model-level bootstrap intervals and paired differences, and add a model-assisted estimator and its ablations.

Model-assisted estimation and prediction-powered inference. Estimating a finite-population mean from a probability sample, using a model fitted elsewhere only to reduce variance while the design supplies validity, is the model-assisted (difference-estimator) approach of survey sampling [Särndal et al., 1992, Breidt and Opsomer, 2017]. Prediction-powered inference [Angelopoulos et al., 2023] is a closely related construction for settings in which model predictions are available on many items and labels on few, and stratified variants have been used to evaluate language models [Fisch et al., 2024]. In our setting the "population" is the benchmark's item set, the "labels" are the target model's 0/1 outcomes on the sampled items, and the "predictions" come from a latent-trait model fitted on the source models. We do not claim the estimator family as new; what we study is how its design (stratification, probability-proportional-to-size allocation) and predictor behave under the protocols above, in particular when targets lie above the range of the source models. Related uses

of sparse or subsampled evaluation include Ghosh et al. [2024], who aggregate sparse measurements across datasets, and Maia Polo et al. [2024b], who estimate performance distributions over prompt variants.

3 Problem Setting and Methods

Setting. A benchmark has N items. We observe a binary matrix $X \in \{0,1\}^{S\times N}$ of correctness of S *source* models on every item. A *target* model t is evaluated on a coreset $I \subset [N]$ of $n \ll N$ items only, giving y_{tj}, $j \in I$; its full-benchmark score is $a_t = \frac{1}{N}\sum_{j=1}^{N} y_{tj}$. A method consists of a *selection* rule that chooses I from X and a seed (never from target data) and a *predictor* $\hat{a}_t$ of a_t from $\{y_{tj}\}_{j\in I}$ and X. Quality is judged against the true a_t of held-out target models: MAE, RMSE, bias, and Spearman and Kendall rank correlations across targets. Every method gets the same coreset size, source/target split and metrics. Hyperparameters are chosen on source models only.

3.1 Reference estimators

Null predicts every target by the mean source score and uses no target items; it shows how much of the task is prior knowledge about the source pool. **Random mean** averages y_{tj} over a uniform random coreset, an unbiased estimator of a_t with variance $\approx a_t(1-a_t)/n$ (up to the finite-population correction). Other baselines are *difficulty-stratified* sampling, *Anchor Points* (weighted) [Vivek et al., 2024], linear *calibration* of the subset mean to the full score learnt on the source models, *ridge* regression of the target's coreset responses on the source models' responses, ***IRT*** prediction with random and with information-selected (greedy D-optimal) items, anchor-selected IRT, and two forms of AIPW (Section 3.2). The full list with the exact role of each (faithful re-implementation, simplified variant or ablation) is in Appendix A. Three published approaches could not be reproduced faithfully with the supplied data and CPU budget and are *not* run: the assessor of Pacchiardi et al. [2024] (needs item embeddings), the learned acquisition policy of Li et al. [2025], and the GAM of metabench [Kipnis et al., 2025]; the information-selected IRT baseline is our own simplified variant and is named as such.

3.2 Model-assisted sampling and difference estimation (MASS)

Predictor. We fit a low-rank logistic latent-trait model on the source models, $\operatorname{logit}\Pr(y_{sj}=1) = r_s + c_j + u_s^\top v_j$ (rank k), by penalised maximum likelihood. Item parameters (c_j, v_j) are then frozen and a target's trait (r_t, u_t) is estimated by MAP from its coreset responses, giving $\hat{p}_{tj}$ for all $j \in [N]$. The default `mass_ens` predictor averages these probabilities (rank $k=4$) with a kernel-weighted average of source models' responses, with weights $\propto \exp(-(d_s - \min_{s'} d_{s'})/\tau)$ where d_s is the Hamming disagreement of source s with the target on the coreset ($\tau = 0.05$). The latent-only variant `mass` uses the first term alone.

Design. Items are ordered by estimated source difficulty $\bar{p}_j$ and sampled by systematic probability-proportional-to-size (PPS) sampling with inclusion probabilities $\pi_j \propto \sqrt{\overline{p_j(1-p_j)}}$, truncated at 1 and summing to n, where the bar denotes the average over source models. Ordering by difficulty acts as implicit stratification; the PPS weights allocate more of the coreset to items whose outcome is uncertain across source models (a Neyman-type allocation for the residual $y_{tj} - \hat{p}_{tj}$).

Estimator. With weights $w_j = 1/\pi_j$ the estimator is the Hájek-form difference estimator

$$\hat{a}_t^{\mathrm{MASS}} = \frac{1}{N}\sum_{j=1}^{N} \hat{p}_{tj} + \frac{\sum_{j\in I} w_j\,(y_{tj} - \hat{p}_{tj})}{\sum_{j\in I} w_j}. \tag{1}$$

This is the model-assisted estimator of survey sampling [Särndal et al., 1992] with the latent-trait model as the assisting model. If $\hat{p}_{tj}$ did not depend on the sampled responses, the correction term would be design-consistent for the mean residual, so that the estimator would be (asymptotically) design-unbiased for a_t whatever the quality of the predictor: a poor predictor then costs variance, not bias, which is the property relevant to extrapolation to stronger models. *Here the predictor* is *fit on the coreset responses, without cross-fitting, so this guarantee holds only approximately; whether the bias*

is negligible in practice is an empirical question that we test (bias under extrapolation, Section 4). The same construction with a uniform random coreset and no PPS is the textbook AIPW/prediction-powered mean (`aipw_rand`); `mass_sel` (design only), `mass_ens_nocorr` (design and predictor, no correction) and `ens_rand_nocorr` (predictor only) isolate the three components. MASS also reports a standard error from the Hájek linearisation (PPS-with-replacement approximation, no finite-population correction); its coverage is checked, not assumed.

4 Experimental Protocol and Results

4.1 Data and protocol

Data. We use the 0/1 item-by-model correctness matrices released with the benchmark-prediction repository of Zhang et al. [2025] (pinned commit `55faff49`, MIT licence): seven Open LLM Leaderboard v2 [Fourrier et al., 2024] benchmarks (ARC-Challenge, BBH, GPQA, IFEval, MATH Lvl 5, MMLU-Pro [Wang et al., 2024], MuSR) and six HELM [Liang et al., 2023] scenarios (commonsense, GSM, LegalBench, MATH, MedQA, MMLU). Item text is not used. Exact duplicate model rows, which occur in the Open LLM files and would place a copy of a target model among the sources, are removed before any split (Appendix B; a with/without-deduplication sensitivity run is reported).

Splits. `random`: 20% of the models, drawn uniformly, are targets (interpolation; 20 repetitions). `extrap`: the top 20% of models by full-benchmark score are targets and all sources are weaker (extrapolation). `family` (Open LLM benchmarks only): all models of randomly chosen organisations are held out; this is coarser than a base-model family and is stated as a limitation. Budgets are $n \in \{10, 20, 50, 100, 200\}$.

Uncertainty. Confidence intervals resample target *models* (bootstrap, B and seeds in Appendix) and are combined with repetition over splits or item-selection seeds; differences between two methods use the same resamples (paired). HELM scenarios share models and are not independent; we do not pool significance across them.

Pre-declared comparison. Before the final sweep we fixed the headline comparison: protocol `extrap`, $n \in \{50, 100\}$, RMSE, MASS versus random mean, versus random + ridge and versus the baseline with the lowest source-only cross-validated RMSE. The method list was frozen before the final sweep; the choice of the `mass_ens` predictor was made after a development pilot on four benchmarks (IFEval, ARC-Challenge, GPQA, GSM), which are flagged and summarised separately. Headline summaries are given on all 13 benchmarks and on the nine non-development ones.

Reading the tables. "RMSE ratio" is the geometric mean over benchmarks of RMSE(method)/RMSE(random mean) at the same n, with a 95% bootstrap interval over benchmarks (the benchmark is the unit; HELM scenarios share models, so these intervals are optimistic for them). Bold means "not distinguishable from the best row of the column" (paired benchmark-level bootstrap), which is not a significance claim against the other rows. Per-benchmark results, with model-level intervals, are in Appendix C. Development benchmarks carry an asterisk.

4.2 Results

Q1: selection plus prediction beats the plain subset mean, but how much depends on the protocol. Tables 1 and 2 and Figure 1 give the answer at fixed budgets. Using only the source models (no target items) is several times worse than the random mean (RMSE ratio 3.9 under extrapolation, 3.1 under interpolation at $n = 50$): the target items carry the information. Under extrapolation, MASS-Ens has an RMSE ratio of 0.79 [0.75, 0.83] to the random mean at $n = 50$ and 0.78 [0.73, 0.83] at $n = 100$, lower on all 13 benchmarks; the latent-only MASS gives 0.85 [0.81, 0.88] and 0.84 [0.79, 0.90]; on the nine non-development benchmarks MASS-Ens gives 0.77 [0.73, 0.80] (Table 1). The textbook AIPW with the latent predictor on random items already gives 0.85 [0.81, 0.88], and the other latent-trait methods fall in the same 0.80–0.85 band, so most of the gain is available without the new design. Under interpolation the ratios are 0.73 [0.70, 0.77] (MASS-Ens) and 0.78 [0.74, 0.82] (MASS) at $n = 50$.

Table 1: Cross-benchmark summary, extrapolation (top 20% of models are targets), $n = 50$ items. Ratios of RMSE to the random-subset mean (1 = no gain, lower is better; τ_b = Kendall rank correlation with the true target scores, mean over benchmarks). *Sources only* uses no target items.

Method	RMSE ratio, all 13	RMSE ratio, 9 non-dev.	wins (of 13)	Kendall τ_b
Sources only	3.89 [2.71, 5.51]	3.80 [2.65, 5.35]	0	– (constant)
Random mean	1.00 [1.00, 1.00]	1.00 [1.00, 1.00]	–	**0.47 [0.40, 0.55]**
Stratified	0.89 [0.86, 0.92]	0.88 [0.85, 0.91]	12	**0.46 [0.39, 0.54]**
Calibrated random	1.05 [0.97, 1.14]	1.06 [0.96, 1.17]	3	**0.47 [0.40, 0.55]**
Random + ridge	1.41 [1.18, 1.68]	1.43 [1.16, 1.74]	3	0.33 [0.26, 0.41]
Anchor points	1.03 [0.80, 1.31]	**0.98 [0.73, 1.32]**	6	0.41 [0.33, 0.48]
IRT (random)	0.84 [0.80, 0.87]	0.83 [0.78, 0.87]	13	0.45 [0.38, 0.53]
IRT (D-opt.)	**0.89 [0.72, 1.10]**	**0.84 [0.66, 1.04]**	8	**0.42 [0.33, 0.52]**
Anchor + IRT	**0.80 [0.76, 0.83]**	**0.79 [0.74, 0.83]**	13	**0.49 [0.43, 0.55]**
AIPW	0.85 [0.81, 0.88]	0.84 [0.79, 0.88]	13	0.45 [0.38, 0.53]
AIPW + ridge (cross-fit)	0.95 [0.89, 1.03]	0.95 [0.88, 1.05]	10	0.43 [0.37, 0.51]
PPI	0.85 [0.81, 0.89]	0.84 [0.79, 0.89]	13	0.45 [0.38, 0.53]
AIPW-kNN	0.85 [0.79, 0.93]	**0.84 [0.76, 0.93]**	11	**0.46 [0.39, 0.54]**
AIPW-Ens	**0.82 [0.78, 0.86]**	**0.81 [0.75, 0.86]**	13	0.46 [0.38, 0.53]
MASS (latent)	0.85 [0.81, 0.88]	0.84 [0.81, 0.87]	12	0.44 [0.38, 0.52]
MASS-Ens	**0.79 [0.75, 0.83]**	**0.77 [0.73, 0.80]**	13	0.46 [0.40, 0.53]

Table 2: As Table 1, interpolation (20% random models are targets), $n = 50$.

Method	RMSE ratio, all 13	RMSE ratio, 9 non-dev.	wins (of 13)	Kendall τ_b
Sources only	2.14 [1.49, 3.00]	2.12 [1.49, 2.92]	2	– (constant)
Random mean	1.00 [1.00, 1.00]	1.00 [1.00, 1.00]	–	**0.73 [0.64, 0.80]**
Stratified	0.84 [0.82, 0.86]	0.82 [0.80, 0.85]	13	0.72 [0.63, 0.79]
Calibrated random	0.75 [0.67, 0.81]	**0.76 [0.69, 0.82]**	13	**0.73 [0.64, 0.80]**
Random + ridge	**0.68 [0.57, 0.79]**	**0.70 [0.59, 0.81]**	13	**0.76 [0.71, 0.80]**
Anchor points	0.89 [0.80, 1.00]	0.86 [0.76, 0.98]	9	**0.75 [0.64, 0.83]**
IRT (random)	0.82 [0.79, 0.84]	0.81 [0.79, 0.84]	13	0.71 [0.63, 0.79]
IRT (D-opt.)	1.13 [0.96, 1.33]	1.18 [0.96, 1.48]	5	**0.69 [0.59, 0.78]**
Anchor + IRT	0.80 [0.76, 0.83]	**0.79 [0.75, 0.83]**	13	**0.73 [0.65, 0.80]**
AIPW	0.83 [0.80, 0.85]	0.82 [0.80, 0.84]	13	0.71 [0.63, 0.79]
AIPW + ridge (cross-fit)	0.86 [0.83, 0.90]	0.85 [0.81, 0.90]	13	0.71 [0.63, 0.78]
PPI	0.83 [0.80, 0.85]	0.82 [0.79, 0.84]	13	0.71 [0.63, 0.79]
AIPW-kNN	**0.76 [0.73, 0.80]**	**0.76 [0.71, 0.80]**	13	**0.73 [0.64, 0.80]**
AIPW-Ens	0.78 [0.75, 0.81]	**0.78 [0.75, 0.81]**	13	0.72 [0.64, 0.80]
MASS (latent)	**0.78 [0.74, 0.82]**	**0.76 [0.72, 0.80]**	13	**0.73 [0.65, 0.81]**
MASS-Ens	**0.73 [0.70, 0.77]**	**0.72 [0.68, 0.77]**	13	**0.74 [0.66, 0.81]**

Pre-declared headline (H-main). Figure 2 gives the paired comparison fixed before the final sweep, at $n \in \{50, 100\}$. Under extrapolation, MASS-Ens versus random + ridge has RMSE ratio 0.56 [0.48, 0.66] ($n = 50$) and 0.48 [0.41, 0.56] ($n = 100$); versus the baseline with the lowest source-only cross-validated RMSE it has 1.01 [0.93, 1.10] and 0.92 [0.83, 1.05]. The intervals contain 1, so MASS-Ens is *not* distinguishable from the best source-CV baseline. The source-CV choice itself is heterogeneous under extrapolation (Anchor Points on 3 benchmarks, anchor-IRT on 3, ridge on 2, information-selected IRT on 2, three other baselines on one each), and picks ridge on GPQA and MuSR, where ridge is indeed below the random mean under extrapolation (RMSE ratios 0.87 and 0.82 at $n = 50$; on the ten other benchmarks except LegalBench, ridge is above it, 1.14–2.32). Under interpolation, versus ridge the ratios are 1.08 [0.93, 1.29] and 1.04 [0.91, 1.19], and versus the source-CV choice 1.11 [0.97, 1.31] and 1.06 [0.93, 1.22]: no advantage for MASS-Ens, and ridge on the subset is the best point estimate (Table 2). At small budgets the point estimate favours the baseline: versus the source-CV choice MASS-Ens has ratio 1.08 [0.96, 1.21] at $n = 10$ and 1.04 [0.94, 1.14] at $n = 20$ under extrapolation (1.18 [0.98, 1.47] and 1.13 [0.95, 1.38] under interpolation); the intervals include 1, and the ratio falls below 1 only from $n = 100$ (0.92 [0.83, 1.05]) and $n = 200$ (0.84 [0.72, 1.02]), again with intervals that include 1.

Q3: when do methods fail? Extrapolation and the sign of the error. Figure 3 shows true versus predicted scores for three benchmarks under extrapolation. Each point averages the predictions of 20 item-selection seeds, so the panels show the systematic part (bias) of each method, not the error of a single run; the printed RMSE is the per-run RMSE of the tables, not that of the plotted averages. Random + ridge is biased low for target models above the source range (median over benchmarks of the mean signed error at $n = 50$: -0.058; random mean -0.001, MASS-Ens -0.008), flattening toward the source mean, and its RMSE ratio to the random mean is 1.41 [1.18, 1.68] (1.65 at $n = 100$).

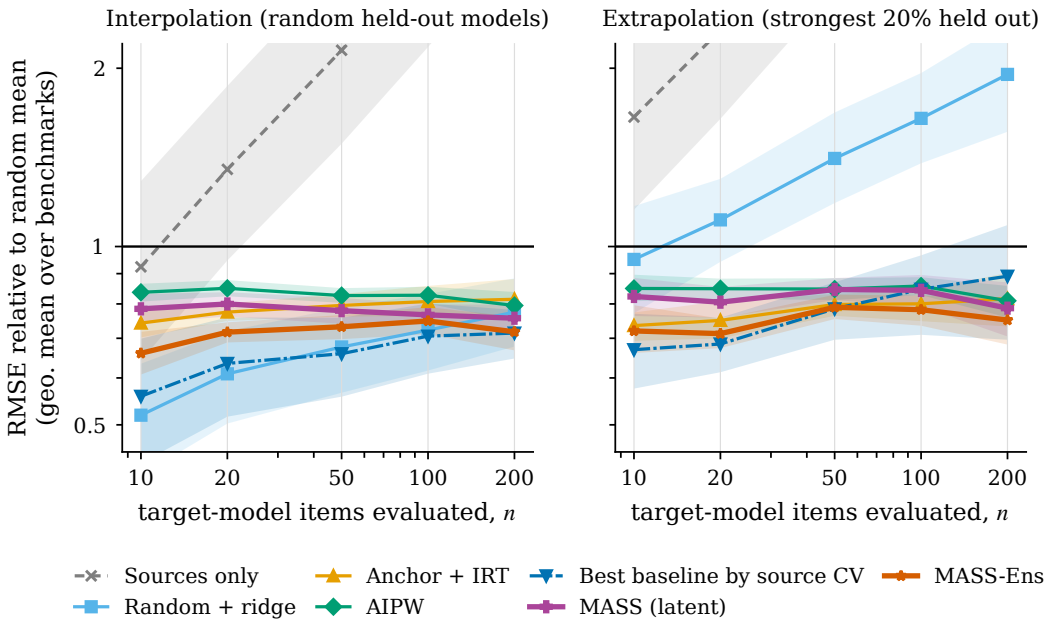


Figure 1: RMSE relative to the random-subset mean as a function of the number of target items, geometric mean over the 13 benchmarks, bands = benchmark-level 95% intervals. Left: interpolation; right: extrapolation. "Best baseline by source CV" selects, per benchmark, the baseline with the lowest RMSE in cross-validation on source models only.

Table 3: Selection × prediction under extrapolation and interpolation, $n = 100$, RMSE ratio to the random-subset mean, geometric mean over benchmarks (10 repetitions; benchmark-level 95% intervals). Columns: predictor; "corr." = difference correction on the sample.

Protocol	Design	sample mean	latent + corr.	ensemble + corr.	ensemble, no corr.
Extrapolation	Random items	1.00 [1.00, 1.00]	0.86 [0.83, 0.89]	0.84 [0.81, 0.87]	1.67 [1.33, 2.09]
	Stratified	0.87 [0.84, 0.91]	0.83 [0.79, 0.89]	–	–
	PPS only	–	0.81 [0.76, 0.87]	–	–
	Stratified + PPS (MASS design)	0.86 [0.81, 0.90]	0.84 [0.79, 0.90]	0.78 [0.74, 0.83]	1.64 [1.31, 2.03]
Interpolation	Random items	1.00 [1.00, 1.00]	0.83 [0.80, 0.86]	0.80 [0.77, 0.83]	0.87 [0.72, 1.05]
	Stratified	0.84 [0.82, 0.87]	0.81 [0.79, 0.84]	–	–
	PPS only	–	0.77 [0.74, 0.80]	–	–
	Stratified + PPS (MASS design)	0.83 [0.79, 0.86]	0.77 [0.74, 0.80]	0.75 [0.71, 0.79]	0.82 [0.69, 0.99]

Anchor Points (-0.026) and calibration (-0.036) show smaller negative biases and ratios of 1.03 [0.80, 1.31] and 1.05 [0.97, 1.14]. The ablations (Table 3) identify the cause of the safety of MASS: with the predictor alone and no difference correction (`mass_ens_nocorr`), the ratio is 1.64 [1.31, 2.03] at $n = 100$, i.e. it inherits the ridge-like bias, whereas the same predictor with the correction gives 0.78. The correction, not the design or the predictor, keeps the estimator from degrading under extrapolation. The design alone (stratified + PPS, Hajek mean) gives 0.86 [0.81, 0.90] and the predictor with correction on random items 0.84 [0.81, 0.87], so each of the three components contributes a little and their combination the most (0.78 [0.74, 0.83]).

Error is not ranking. Lower RMSE does not translate into better ranking of the targets. Under extrapolation the Kendall τ_b of MASS-Ens is 0.46 [0.40, 0.53] against 0.47 [0.40, 0.55] for the random mean at $n = 50$ (paired difference -0.010 [-0.040, 0.037]); random + ridge falls to 0.33 (difference -0.140 [-0.192, -0.093]). Under interpolation MASS-Ens improves τ_b by 0.013 [0.003, 0.025]. A further reason is spread: with strong models as targets, the median over benchmarks of RMSE divided by the standard deviation of the true target scores is 1.11 for MASS-Ens (1.42 for the random mean) under extrapolation, against 0.33 (0.46) under interpolation: the errors are as large as the differences between the models one wants to rank. On the low-spread benchmarks (GPQA,

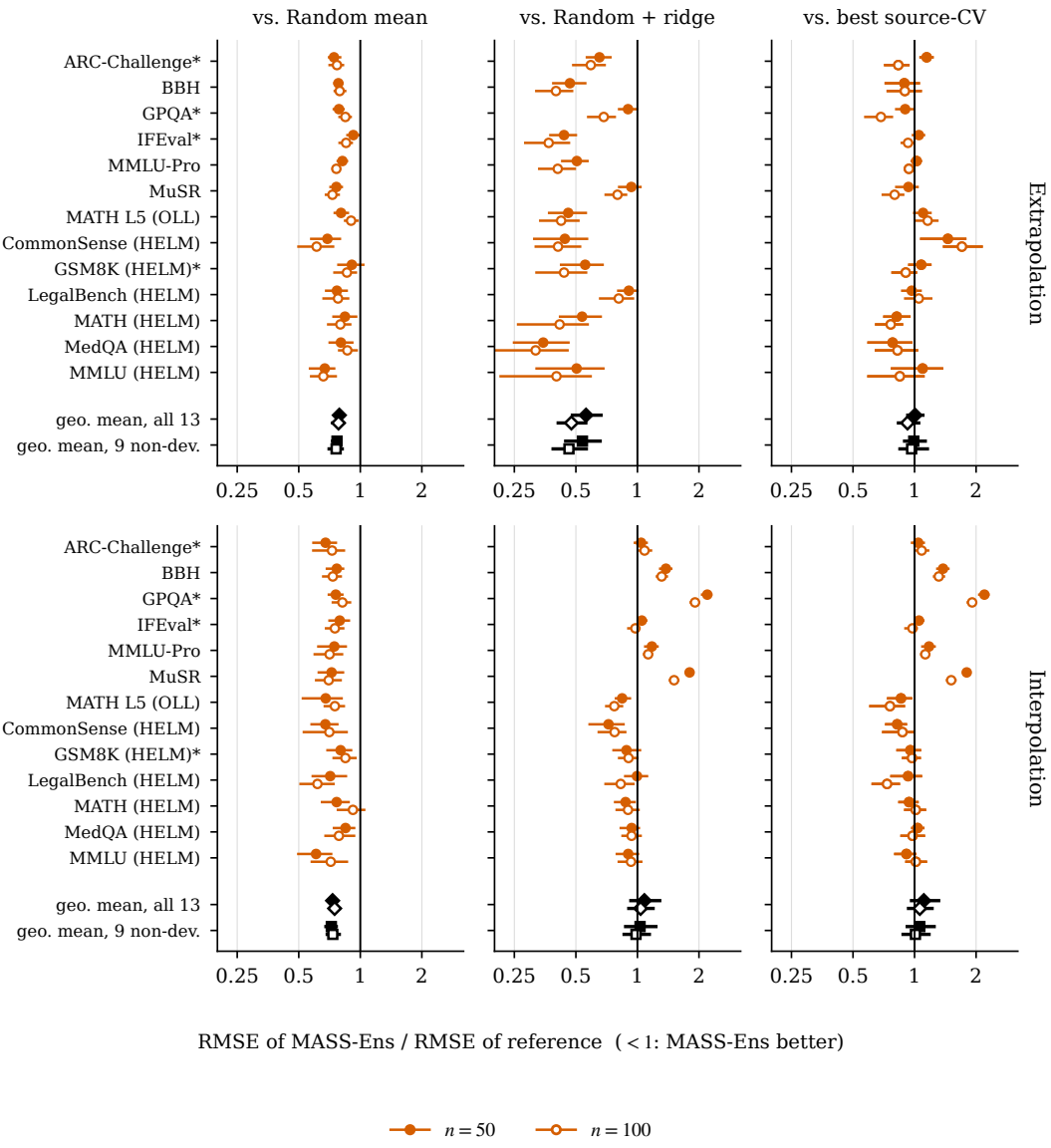


Figure 2: Pre-declared comparison. Per benchmark, RMSE(MASS-Ens)/RMSE(reference) with paired model-level (extrapolation) or repetition-level (interpolation) 95% intervals, for $n = 50$ (filled) and $n = 100$ (open); values below 1 favour MASS-Ens. Bottom rows: geometric mean over all 13 benchmarks and over the nine non-development ones. The latent-only MASS is in Figure 4 (Appendix).

MuSR, MATH Lvl 5) rankings are no better than with the random mean (e.g. τ_b at $n = 50$ for GPQA/MuSR: 0.29/0.27 for MASS-Ens and 0.29/0.27 for the random mean under extrapolation).

Effective sample size and budget. Defining the gain as $(\mathrm{RMSE}_{\mathrm{random}}/\mathrm{RMSE}_{\mathrm{method}})^2$ at equal n, the median over benchmarks for MASS-Ens is 1.6 at $n = 50$ (smallest benchmark 1.17) under extrapolation and 1.8 (smallest 1.40) under interpolation (Figure 7, Appendix): the method is worth roughly 60–80% more random items, not an order of magnitude. This is the practical size of the effect.

Does the reported uncertainty hold? MASS-Ens reports a standard error. Averaged over benchmarks, the nominal 95% interval covers the true target score for 74% (interpolation: 77%) of targets at $n = 10$, 87% at $n = 20$, 93% at $n = 50$ (extrapolation) and 92% (interpolation), 95–96% at

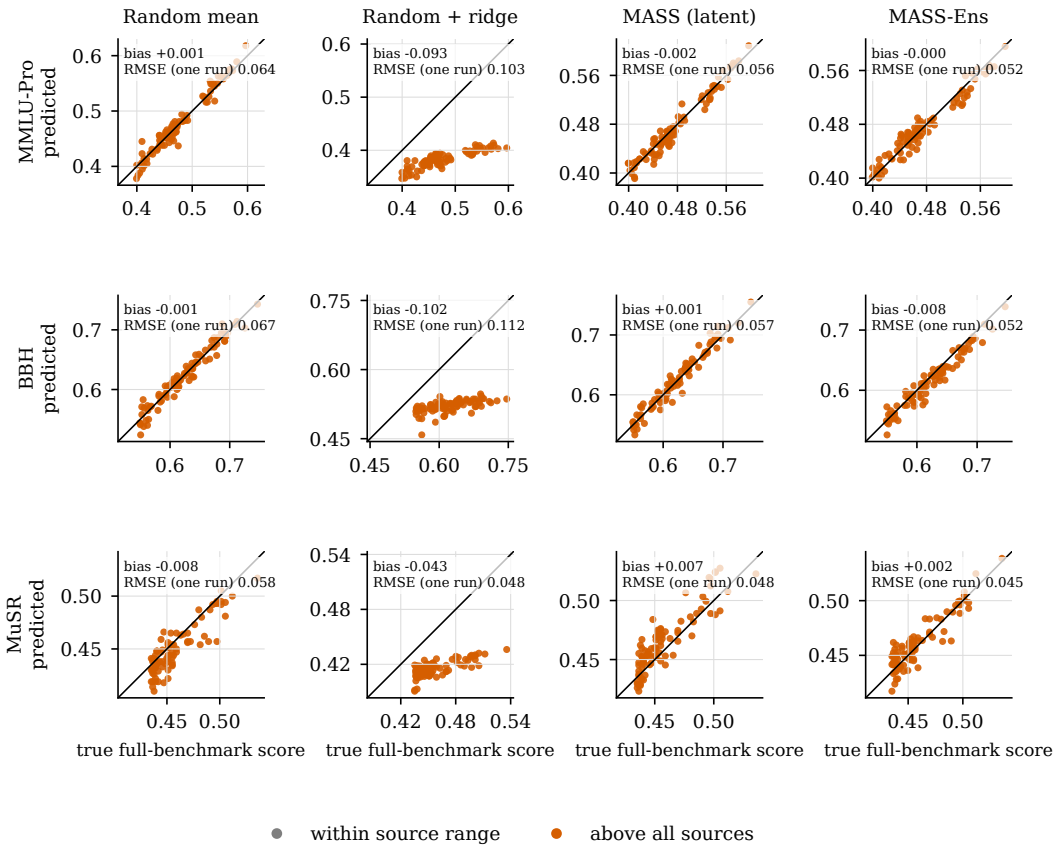


Figure 3: Extrapolation, $n = 50$: true versus predicted full-benchmark score of each target model (average over 20 item-selection seeds) for three benchmarks and four methods; points above the source range are orange. Points are averages over seeds and show the systematic part of the error (bias); the printed RMSE is the per-run RMSE of the tables (fractions, not points).

$n = 100$ and 97% at $n = 200$ (Figure 8, Appendix). It under-covers for $n \leq 20$, plausibly because the predictor is fit on the sampled items and the standard error ignores this (Section 3.2), which we did not test separately; we therefore do not claim valid intervals at small n.

Q3: sensitivity. Table 4 and Figure 9 (Appendix) vary what the protocol fixes. Holding out whole organisations (a coarse family split) behaves like interpolation: ridge again has the lower point estimate (0.66 [0.54, 0.80] versus 0.75 [0.71, 0.79] for MASS-Ens; the intervals overlap) on the seven Open LLM benchmarks. With fewer source models or a larger held-out fraction, ridge degrades visibly under extrapolation while the correction-based estimators change little. Keeping duplicate model rows moves extrapolation ratios (MASS-Ens 0.78 to 0.86, AIPW 0.86 to 0.90) while leaving their ordering with respect to ridge unchanged, so duplicate removal matters for the magnitude of the reported gains.

Prior-work hypotheses. The code repository of Zhang et al. [2025] states three conclusions, which we treated as hypotheses. (i) *Random sampling plus regression is a strong baseline*: confirmed under interpolation (ridge has the best point estimate, ratio 0.68 at $n = 50$) and refuted under extrapolation (1.41). (ii) *Methods degrade when targets exceed the source models*: confirmed for ridge, and mildly for Anchor Points and calibration; qualified for estimators with a difference correction on the sample, which stay below the random mean's RMSE (0.79–0.85) but do not improve ranking. (iii) *AIPW-type estimators help*: confirmed for RMSE against the random mean (13/13 benchmarks) with a latent-trait predictor; with a ridge predictor and cross-fitted residuals the ratio is 0.95 [0.89, 1.03], so the benefit depends on the predictor.

Table 4: Robustness of RMSE ratio to the random mean ($n = 100$, 10 repetitions, geometric mean over the benchmarks run). "dups kept": exact duplicate model rows not removed. Family-out: Open LLM benchmarks only.

Method	Random (13)	Random, dups kept (13)	Family-out (7)	Extrap. (13)	Extrap., dups kept (13)
Sources only	3.13 [2.17, 4.36]	3.26 [2.25, 4.59]	2.34 [1.46, 3.69]	5.63 [3.92, 7.99]	5.90 [4.07, 8.43]
Random + ridge	0.72 [0.62, 0.83]	0.74 [0.63, 0.86]	0.66 [0.54, 0.80]	1.65 [1.38, 1.96]	1.69 [1.42, 1.99]
Anchor + IRT	0.81 [0.76, 0.86]	0.83 [0.78, 0.89]	0.79 [0.73, 0.84]	0.80 [0.75, 0.84]	0.85 [0.81, 0.89]
AIPW	0.83 [0.80, 0.86]	0.85 [0.82, 0.87]	0.83 [0.79, 0.87]	0.86 [0.83, 0.89]	0.90 [0.87, 0.93]
AIPW-Ens	0.80 [0.77, 0.83]	0.81 [0.78, 0.84]	0.80 [0.76, 0.84]	0.84 [0.81, 0.87]	0.88 [0.84, 0.92]
MASS (latent)	0.77 [0.74, 0.80]	0.78 [0.75, 0.82]	0.79 [0.74, 0.84]	0.84 [0.79, 0.90]	0.85 [0.79, 0.91]
MASS-Ens	0.75 [0.71, 0.79]	0.75 [0.72, 0.79]	0.75 [0.71, 0.79]	0.78 [0.74, 0.83]	0.86 [0.80, 0.92]

5 Discussion and Limitations

What the evidence supports. Under a common protocol, target items plus source-model information reduce the error of the full-score estimate by about a fifth to a quarter relative to a random subset, and the correction term makes this robust to extrapolation. It does not show that MASS is better than existing methods in general: against the baseline chosen by source-only cross-validation the difference is within noise, anchor-IRT is not distinguishable from MASS-Ens in the extrapolation headline table, and under interpolation ridge on a random subset is as good or better. Ranking is not improved, and under extrapolation the error is of the size of the spread among the strongest models.

Limitations. (i) *Development choice.* The ensemble predictor of MASS-Ens was chosen after a pilot that looked at target-model errors on one split of IFEval, ARC-Challenge, GPQA and GSM; those benchmarks are flagged, the latent-only MASS is reported alongside, and the nine non-development benchmarks give a similar summary. The target-side results of the baselines existed before the source-CV baseline selection was computed (the selection itself uses only sources). The method list was frozen before the final sweep; later runs (sensitivity) use a reduced method set and 10 repetitions. (ii) *One extrapolation split per benchmark*: the strongest 20% of models; uncertainty comes from model-level bootstrap and item-selection seeds, not from alternative splits. Targets in the top 20% are few for HELM benchmarks (16–17 models). The family-out split uses the organisation prefix of model names, coarser than a base-model family, and HELM has no family analogue. (iii) *Dependence*: the six HELM scenarios share models, and benchmark-level intervals treat the benchmarks as independent; we pool no significance statements and report "wins" without multiplicity correction. The 13 benchmarks are the unit of the cross-benchmark summaries; bootstrap $B = 200$ for model-level intervals. (iv) *Estimator.* The predictor is fit on the sampled items without cross-fitting, so design-unbiasedness is only approximate; the standard error ignores this and under-covers for $n \leq 20$. (v) *Scope.* Binary scores only; no item text is used; Pacchiardi et al.'s assessor, Active Evaluation Acquisition and the metabench GAM were not run (they need item embeddings, a trained policy or specialised fitting not available here), and `anchor_irt`, `anchor_aipw` and `irt_info` are simplified variants under their own names, so their results do not speak for the original methods. Low-spread benchmarks (GPQA, MuSR, MATH Lvl 5) give poor rankings for every method; we report them instead of dropping them. Source models are open-weight leaderboard models with many near-duplicates and fine-tunes, which makes interpolation easier than evaluating a new architecture. (vi) *Compute.* Every run is CPU only; the heaviest sweep (MMLU-Pro, 12k items) takes about 15 h of wall time per method group on a loaded 2-CPU host (Appendix E).

Broader impact. The work concerns cheaper evaluation of language models; the risk is over-trusting a subset-based score, in particular for ranking or for models unlike the source pool. We recommend reporting the interval and the source pool, and using the full benchmark for decisions about the strongest models.

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A Method list

All methods are our own implementations (`code/src/methods*.py`); the released `benchpred` package is not used. Selection and all fitting use source models only. Roles: P = proposed, B = baseline (faithful re-implementation of the textbook or published form), V = simplified variant under its own name, A = ablation, R = reference row.

Label (paper)	code name	role	description
Sources only	`sources_only`	R	mean source-model score, no target items
Random mean	`random`	R/B	mean over a uniform random coreset
Stratified	`strat_random`	B	systematic sampling along source difficulty, sample mean
Calibrated random	`calib`	B	linear map from subset mean to full score learnt on sources
Random + ridge	`ridge`	B	ridge regression of target subset responses on source responses, random items
Anchor points	`anchor`	B	k-medoid anchors on $1-$correlation, cluster-size weights [Vivek et al., 2024]
IRT (random)	`irt_rand`	B	rank-2 logistic latent-trait prediction, random items
IRT (D-opt.)	`irt_info`	V	greedy D-optimal items, same predictor (not metabench)
Anchor + IRT	`anchor_irt`	V	anchors plus latent-trait prediction (tinyBenchmarks-style, simplified)
AIPW	`aipw_rand`	B	textbook: mean prediction over all items + mean residual on sample, latent predictor
AIPW + ridge (cross-fit)	`aipw_ridge_cf`	B	same with ridge predictor and 5-fold cross-fitted residuals
(AIPW, repository variant)	`aipw_ridge_repo`	V	ridge(α=10) with in-sample residual$/(1 + n/N_{\text{rest}})$, as in the data repository
PPI	`ppi`	B	prediction-powered mean with tuned weight $\lambda \in [0, 1]$
AIPW-kNN	`aipw_knn`	V	kernel-weighted source-model predictor, random items
AIPW-Ens	`aipw_ens`	A	ensemble predictor with correction, random items
MASS (latent)	`mass`	P	MASS design, latent predictor, Hajek correction
MASS-Ens	`mass_ens`	P	MASS design, ensemble predictor, Hajek correction
(MASS ablations)	`mass_sel`, `mass_ens_nocorr`, `ens_rand_nocorr`, `aipw_strat`, `aipw_pps`, `mass_k1`, `mass_k4`, `aipw_k1`, `aipw_k4`	A	design only; design + predictor without correction; predictor only; stratification only; PPS only; latent rank 1/4

Not run: the assessor of Pacchiardi et al. [2024], the acquisition policy of Li et al. [2025], the GAM of Kipnis et al. [2025].

B Data and protocol details

Table 5: Score matrices after loading (raw models, models after removing exact duplicate rows, items, accuracy range across models, items solved by all or by no model). * = development benchmark. Models with near-zero accuracy are dropped from MATH Lvl 5 (394 raw rows remain).

Benchmark	Suite	Models raw	(dedup.)	Items	Model accuracy (%) min	median	max	const. items
ARC-Challenge*	OLL	218	209	1172	20.5	55.7	75.1	32
BBH	OLL	446	425	5761	24.1	44.2	74.7	0
GPQA*	OLL	447	424	1192	22.5	28.2	43.7	0
IFEval*	OLL	448	427	541	3.7	27.7	88.0	7
MMLU-Pro	OLL	448	425	12032	10.3	27.8	59.7	1
MuSR	OLL	448	427	756	32.3	39.9	53.6	10
MATH L5 (OLL)	OLL	394	374	894	0.1	2.7	35.6	119
CommonSense (HELM)	HELM	83	82	500	22.2	87.5	97.2	0
GSM8K (HELM)*	HELM	83	83	1000	2.8	73.4	95.6	3
LegalBench (HELM)	HELM	79	79	2047	22.8	56.8	70.2	53
MATH (HELM)	HELM	83	83	437	2.5	65.0	93.4	1
MedQA (HELM)	HELM	83	82	1000	21.5	60.9	86.9	2
MMLU (HELM)	HELM	83	82	567	22.9	63.7	81.1	1

Latent-trait model: rank k (2 for `mass`, `irt_rand`; 4 for `mass_ens`), penalised maximum likelihood with Adam (300 steps, learning rate 0.05, seed 0), target trait by MAP on the coreset. Item-selection seed of repetition r at budget n is $10000r + n$; the random split of repetition r uses seed $1000 + r$

(family: $2000 + r$). Bootstrap: percentile, $B = 200$ for model-level intervals, 2000 resamples of benchmarks for cross-benchmark summaries. 20 repetitions in the main grid, 10 in sensitivity runs.

C Per-benchmark results

RMSE in accuracy points ($\times 100$) with model-level (extrapolation) or repetition-level (interpolation) 95% intervals; bold = best point estimate in the row; † = paired interval of (method − MASS-Ens) excludes 0; $*$ = development benchmark. Kendall τ_b (bold = best point estimate; † as above):

Table 6: RMSE, extrapolation, $n = 50$.

Benchmark	Sources only	Random mean	Random + ridge	Anchor + IRT	AIPW	MASS (latent)	MASS-Ens
ARC-Challenge*	18.4 [17.5, 19.3]†	5.5 [5.2, 5.7]†	6.2 [5.6, 6.8]†	4.2 [3.9, 4.4]	4.4 [4.1, 4.6]†	4.1 [3.8, 4.2]	**4.1 [3.8, 4.3]**
BBH	22.5 [21.3, 23.5]†	6.7 [6.5, 6.8]†	11.2 [10.2, 12.1]†	5.4 [5.1, 5.6]	5.8 [5.6, 5.9]†	5.7 [5.4, 5.9]†	**5.2 [5.1, 5.4]**
GPQA*	8.1 [7.4, 8.7]†	6.9 [6.6, 7.1]†	6.0 [5.4, 6.6]†	5.7 [5.5, 5.9]†	6.4 [6.1, 6.5]†	6.2 [5.9, 6.5]†	**5.4 [5.2, 5.6]**
IFEval*	41.9 [40.1, 43.7]†	5.7 [5.5, 5.9]†	12.1 [11.3, 12.9]†	**5.0 [4.8, 5.2]**	5.3 [5.1, 5.5]	5.8 [5.5, 6.0]†	5.3 [5.0, 5.5]
MMLU-Pro	24.9 [23.9, 26.0]†	6.4 [6.2, 6.5]†	10.3 [9.7, 11.1]†	5.8 [5.6, 6.0]†	5.6 [5.4, 5.8]†	5.6 [5.5, 5.8]†	**5.2 [5.0, 5.4]**
MuSR	7.7 [7.2, 8.4]†	5.8 [5.7, 6.0]†	4.8 [4.4, 5.3]	4.6 [4.4, 4.8]	5.0 [4.8, 5.2]†	4.8 [4.6, 5.1]†	**4.5 [4.2, 4.7]**
MATH L5 (OLL)	16.1 [14.6, 17.5]†	5.2 [5.0, 5.5]†	9.2 [8.1, 10.2]†	**3.8 [3.4, 4.2]**	4.4 [4.1, 4.6]	4.5 [4.2, 4.8]†	4.2 [3.9, 4.5]
CommonSense (HELM)	16.7 [16.2, 17.3]†	2.3 [2.1, 2.4]†	3.7 [3.2, 4.1]†	1.7 [1.5, 1.8]	2.2 [2.0, 2.4]†	2.0 [1.7, 2.1]†	**1.6 [1.4, 1.8]**
GSM8K (HELM)*	32.7 [31.2, 33.9]†	3.4 [2.9, 3.7]	5.5 [5.1, 5.9]†	2.8 [2.4, 3.1]	2.8 [2.5, 3.0]	**2.7 [2.4, 2.9]**	3.0 [2.6, 3.4]
LegalBench (HELM)	15.3 [13.9, 16.6]†	6.0 [5.4, 6.2]†	5.0 [4.4, 5.5]†	4.7 [4.3, 5.1]	**4.4 [3.9, 4.7]**	4.8 [4.1, 5.4]	4.6 [4.1, 4.8]
MATH (HELM)	38.3 [36.7, 39.7]†	4.1 [3.8, 4.4]†	6.5 [5.8, 7.0]†	**3.3 [2.9, 3.6]**	3.3 [3.0, 3.6]	3.3 [3.0, 3.5]	3.5 [3.0, 3.8]
MedQA (HELM)	27.4 [25.5, 28.9]†	4.8 [4.4, 5.1]†	11.1 [9.8, 12.2]†	4.2 [3.8, 4.6]	4.2 [3.9, 4.4]	4.4 [4.0, 4.7]†	**3.9 [3.5, 4.1]**
MMLU (HELM)	18.0 [16.5, 19.6]†	5.7 [5.4, 6.0]†	7.6 [6.3, 8.9]†	4.0 [3.4, 4.4]	4.2 [3.9, 4.4]	4.3 [3.5, 4.9]	**3.8 [3.4, 4.1]**

Table 7: RMSE, extrapolation, $n = 100$.

Benchmark	Sources only	Random mean	Random + ridge	Anchor + IRT	AIPW	MASS (latent)	MASS-Ens
ARC-Challenge*	18.4 [17.4, 19.2]†	3.8 [3.6, 3.9]†	5.0 [4.4, 5.4]†	3.0 [2.8, 3.1]	3.1 [2.9, 3.2]	**2.8 [2.7, 3.0]**	2.9 [2.7, 3.1]
BBH	22.5 [21.7, 23.4]†	4.6 [4.4, 4.7]†	9.1 [8.4, 9.8]†	4.0 [3.9, 4.2]†	4.0 [3.8, 4.1]†	3.9 [3.7, 4.0]†	**3.6 [3.5, 3.7]**
GPQA*	8.1 [7.6, 8.7]†	4.6 [4.4, 4.7]†	5.6 [5.1, 6.1]†	4.0 [3.8, 4.1]	4.4 [4.2, 4.5]†	4.0 [3.8, 4.1]	**3.9 [3.7, 4.0]**
IFEval*	41.9 [39.5, 43.7]†	3.9 [3.7, 4.0]†	8.9 [8.0, 9.7]†	3.5 [3.4, 3.7]†	**3.3 [3.2, 3.4]**	3.9 [3.6, 4.2]†	3.3 [3.1, 3.5]
MMLU-Pro	24.9 [23.8, 25.9]†	4.8 [4.7, 4.9]†	9.0 [8.2, 9.7]†	4.2 [4.0, 4.3]†	3.9 [3.7, 4.0]†	4.0 [3.9, 4.1]†	**3.7 [3.6, 3.8]**
MuSR	7.7 [7.2, 8.3]†	4.3 [4.1, 4.4]†	3.9 [3.6, 4.2]†	3.4 [3.2, 3.5]†	3.5 [3.4, 3.6]†	3.3 [3.1, 3.4]	**3.1 [2.9, 3.3]**
MATH L5 (OLL)	16.1 [14.7, 17.5]†	3.4 [3.2, 3.5]†	7.2 [6.2, 7.9]†	**2.6 [2.3, 2.9]**†	3.0 [2.9, 3.2]	3.2 [3.0, 3.4]†	3.0 [2.8, 3.3]
CommonSense (HELM)	16.7 [16.2, 17.3]†	1.8 [1.7, 1.9]†	2.7 [2.5, 2.9]†	**1.1 [0.9, 1.2]**	1.6 [1.5, 1.7]†	1.2 [1.1, 1.4]	1.1 [0.9, 1.3]
GSM8K (HELM)*	32.7 [31.3, 33.8]†	2.3 [2.1, 2.5]†	4.6 [4.2, 5.1]†	1.9 [1.6, 2.2]	2.2 [1.9, 2.4]	**1.9 [1.7, 2.1]**	2.0 [1.8, 2.2]
LegalBench (HELM)	15.3 [14.1, 16.6]†	4.0 [3.8, 4.2]†	3.9 [3.3, 4.3]†	**3.0 [2.7, 3.3]**	3.4 [3.1, 3.6]	3.3 [2.9, 3.5]	3.1 [2.7, 3.5]
MATH (HELM)	38.3 [36.5, 39.9]†	2.6 [2.3, 2.8]†	5.0 [4.1, 5.7]†	2.2 [1.9, 2.4]	2.3 [2.1, 2.5]	2.6 [2.3, 2.9]†	**2.1 [1.7, 2.3]**
MedQA (HELM)	27.4 [25.6, 29.2]†	3.6 [3.3, 3.7]†	9.7 [8.3, 10.9]†	**3.0 [2.7, 3.1]**	3.3 [3.1, 3.4]	3.2 [2.7, 3.5]	3.1 [2.8, 3.3]
MMLU (HELM)	18.0 [16.6, 19.5]†	3.6 [3.4, 3.7]†	5.9 [4.7, 6.9]†	2.6 [2.2, 2.8]	2.7 [2.5, 2.8]	2.9 [2.2, 3.6]	**2.4 [2.1, 2.6]**

Table 8: RMSE, interpolation, $n = 50$.

Benchmark	Sources only	Random mean	Random + ridge	Anchor + IRT	AIPW	MASS (latent)	MASS-Ens
ARC-Challenge*	12.9 [12.4, 13.4]†	5.8 [5.3, 6.4]†	**3.8 [3.5, 4.0]**	4.1 [3.9, 4.3]	4.4 [4.1, 4.7]†	4.2 [4.0, 4.4]†	3.9 [3.7, 4.2]
BBH	11.7 [11.4, 11.9]†	6.9 [6.5, 7.3]†	**3.8 [3.6, 4.1]**†	5.3 [5.2, 5.5]	5.7 [5.5, 6.0]†	5.5 [5.3, 5.8]	5.3 [5.1, 5.4]
GPQA*	3.7 [3.6, 3.9]†	6.4 [6.0, 6.7]†	**2.2 [2.1, 2.3]**†	5.7 [5.5, 5.8]†	5.8 [5.6, 6.0]†	5.6 [5.4, 5.8]†	4.8 [4.6, 5.0]
IFEval*	20.3 [19.8, 20.8]†	5.4 [5.0, 5.8]†	**4.1 [3.9, 4.2]**†	4.3 [4.2, 4.4]	4.5 [4.3, 4.6]	4.2 [4.1, 4.4]	4.3 [4.2, 4.4]
MMLU-Pro	13.0 [12.7, 13.3]†	5.7 [5.2, 6.4]†	**3.6 [3.5, 3.9]**†	4.5 [4.4, 4.7]†	4.8 [4.6, 4.9]†	4.7 [4.6, 4.9]†	4.3 [4.1, 4.4]
MuSR	4.0 [3.9, 4.1]†	6.2 [5.6, 6.7]†	**2.5 [2.4, 2.6]**†	4.7 [4.6, 4.9]†	5.1 [4.8, 5.3]†	4.7 [4.6, 4.9]†	4.5 [4.3, 4.6]
MATH L5 (OLL)	7.0 [6.8, 7.2]†	3.2 [2.8, 3.7]†	2.6 [2.4, 2.7]†	2.3 [2.2, 2.4]	2.7 [2.5, 2.9]†	**2.1 [2.0, 2.2]**	2.2 [2.1, 2.3]
CommonSense (HELM)	15.5 [14.2, 16.6]†	4.5 [4.1, 4.9]†	4.2 [3.6, 4.9]†	3.9 [3.5, 4.3]†	3.8 [3.4, 4.3]†	3.4 [3.0, 3.8]	**3.0 [2.7, 3.3]**
GSM8K (HELM)*	24.6 [23.3, 25.8]†	5.2 [4.8, 5.6]†	4.7 [4.4, 5.1]	4.4 [4.0, 4.8]	4.5 [4.3, 4.7]	4.8 [4.5, 5.2]†	**4.2 [3.8, 4.6]**
LegalBench (HELM)	10.9 [10.2, 11.6]†	6.0 [5.2, 6.7]†	4.3 [3.9, 4.7]	4.8 [4.3, 5.3]	4.5 [4.1, 4.9]	4.4 [4.1, 4.8]	**4.3 [3.8, 4.6]**
MATH (HELM)	27.3 [26.3, 28.2]†	5.5 [5.0, 6.3]†	4.8 [4.6, 5.1]†	4.9 [4.6, 5.3]†	4.7 [4.3, 5.3]	4.5 [4.2, 4.7]	**4.2 [3.9, 4.6]**
MedQA (HELM)	15.9 [14.9, 16.7]†	5.8 [5.4, 6.3]†	5.3 [5.0, 5.6]	4.8 [4.5, 5.1]	5.0 [4.6, 5.4]	**4.7 [4.4, 5.0]**	4.9 [4.6, 5.3]
MMLU (HELM)	11.7 [10.8, 12.5]†	6.3 [5.5, 7.1]†	4.3 [3.8, 4.7]	4.5 [4.2, 4.8]†	4.8 [4.4, 5.3]†	4.3 [3.9, 4.5]†	**3.8 [3.5, 4.2]**

Table 9: RMSE, interpolation, $n = 100$.

Benchmark	Sources only	Random mean	Random + ridge	Anchor + IRT	AIPW	MASS (latent)	MASS-Ens
ARC-Challenge*	12.9 [12.4, 13.6]†	4.0 [3.6, 4.6]†	**2.7 [2.5, 2.8]**†	2.9 [2.7, 3.1]	3.1 [3.0, 3.3]†	2.8 [2.7, 2.9]	2.9 [2.8, 3.0]
BBH	11.7 [11.4, 11.9]†	4.9 [4.6, 5.3]†	**2.7 [2.6, 2.9]**†	3.6 [3.5, 3.8]	4.0 [3.9, 4.2]†	3.9 [3.7, 4.1]†	3.6 [3.5, 3.7]
GPQA*	3.7 [3.6, 3.8]†	4.3 [4.0, 4.7]†	**1.8 [1.8, 1.9]**†	3.9 [3.7, 4.0]†	3.8 [3.7, 4.0]†	3.7 [3.6, 3.9]†	3.5 [3.4, 3.7]
IFEval*	20.3 [19.7, 20.7]†	3.7 [3.4, 4.0]†	2.8 [2.7, 3.0]	3.0 [2.9, 3.1]†	3.0 [2.9, 3.1]†	2.8 [2.7, 2.9]	**2.8 [2.7, 2.9]**
MMLU-Pro	13.0 [12.6, 13.3]†	4.3 [3.8, 4.8]†	**2.7 [2.6, 2.8]**†	3.3 [3.2, 3.5]†	3.5 [3.3, 3.6]†	3.2 [3.0, 3.3]	3.1 [2.9, 3.2]
MuSR	4.0 [3.9, 4.1]†	4.3 [4.0, 4.6]†	**2.0 [1.9, 2.1]**†	3.2 [3.0, 3.3]	3.4 [3.2, 3.5]†	3.2 [3.1, 3.4]†	3.0 [2.9, 3.1]
MATH L5 (OLL)	7.0 [6.8, 7.2]†	2.0 [1.8, 2.1]†	1.9 [1.8, 2.1]†	**1.5 [1.4, 1.6]**	1.8 [1.7, 1.9]†	1.5 [1.4, 1.6]	1.5 [1.4, 1.6]
CommonSense (HELM)	15.5 [14.2, 16.8]†	3.0 [2.6, 3.4]†	2.7 [2.4, 3.1]†	2.9 [2.6, 3.2]†	2.4 [2.1, 2.7]†	2.3 [2.1, 2.5]	**2.1 [1.9, 2.3]**
GSM8K (HELM)*	24.6 [23.6, 26.0]†	3.8 [3.3, 4.3]†	3.5 [3.2, 3.9]†	3.4 [3.1, 3.6]	3.3 [3.0, 3.6]	**3.2 [2.9, 3.4]**	3.2 [3.0, 3.4]
LegalBench (HELM)	10.9 [10.2, 11.5]†	4.6 [4.1, 5.0]†	3.4 [3.1, 3.8]†	3.2 [3.0, 3.5]†	3.5 [3.2, 3.7]†	3.2 [3.0, 3.4]†	**2.8 [2.6, 3.1]**
MATH (HELM)	27.3 [26.3, 28.2]†	3.4 [3.0, 4.0]	3.5 [3.2, 3.8]	3.3 [3.0, 3.5]	3.1 [2.9, 3.4]	**2.9 [2.8, 3.1]**	3.1 [2.9, 3.3]
MedQA (HELM)	15.9 [14.9, 16.7]†	4.1 [3.7, 4.6]†	3.5 [3.1, 3.7]	3.6 [3.3, 4.0]	3.6 [3.3, 3.8]	3.3 [3.0, 3.5]	**3.3 [3.0, 3.5]**
MMLU (HELM)	11.7 [10.7, 12.5]†	3.9 [3.3, 4.4]†	3.0 [2.7, 3.3]	2.9 [2.7, 3.1]	2.9 [2.7, 3.2]	**2.7 [2.4, 2.9]**	2.8 [2.6, 2.9]

Table 10: Kendall τ_b, extrapolation, $n = 50$.

Benchmark	Random mean	Random + ridge	Anchor + IRT	AIPW	MASS (latent)	MASS-Ens
ARC-Challenge*	**0.46 [0.37, 0.53]**	0.30 [0.24, 0.38]†	0.41 [0.34, 0.47]	0.43 [0.35, 0.50]	0.41 [0.35, 0.47]	0.42 [0.34, 0.48]
BBH	**0.51 [0.46, 0.55]**	0.24 [0.19, 0.30]†	0.48 [0.42, 0.54]	0.49 [0.44, 0.54]	0.48 [0.44, 0.53]	0.48 [0.43, 0.53]
GPQA*	0.29 [0.24, 0.32]	0.23 [0.18, 0.28]	0.29 [0.22, 0.36]	0.27 [0.23, 0.31]	**0.30 [0.25, 0.35]**	0.29 [0.25, 0.34]
IFEval*	**0.73 [0.71, 0.76]**†	0.64 [0.59, 0.68]†	0.72 [0.69, 0.75]	0.72 [0.69, 0.75]	0.70 [0.67, 0.73]	0.70 [0.68, 0.73]
MMLU-Pro	**0.52 [0.45, 0.58]**†	0.33 [0.27, 0.38]†	0.51 [0.45, 0.55]	0.50 [0.43, 0.56]	0.49 [0.43, 0.54]	0.48 [0.42, 0.53]
MuSR	0.27 [0.21, 0.33]	0.27 [0.19, 0.33]	0.29 [0.22, 0.36]	0.27 [0.21, 0.32]	**0.30 [0.24, 0.34]**	0.27 [0.20, 0.33]
MATH L5 (OLL)	0.58 [0.51, 0.63]	0.45 [0.38, 0.50]†	0.58 [0.50, 0.65]	0.56 [0.49, 0.61]	0.57 [0.50, 0.61]	**0.59 [0.51, 0.63]**
CommonSense (HELM)	0.22 [0.12, 0.30]†	0.11 [0.03, 0.17]†	**0.46 [0.19, 0.66]**	0.18 [0.10, 0.27]†	0.25 [0.16, 0.32]†	0.46 [0.33, 0.57]
GSM8K (HELM)*	**0.58 [0.46, 0.67]**	0.49 [0.39, 0.57]	0.58 [0.42, 0.73]	0.53 [0.41, 0.63]	0.52 [0.40, 0.65]	0.53 [0.37, 0.65]
LegalBench (HELM)	0.38 [0.23, 0.52]	0.33 [0.25, 0.39]	**0.40 [0.27, 0.53]**	0.37 [0.23, 0.49]	0.26 [0.06, 0.41]	0.31 [0.14, 0.46]
MATH (HELM)	**0.58 [0.41, 0.70]**	0.51 [0.37, 0.64]	0.57 [0.42, 0.71]	0.56 [0.40, 0.69]	0.57 [0.44, 0.65]	0.56 [0.41, 0.69]
MedQA (HELM)	0.56 [0.44, 0.65]	0.27 [0.16, 0.38]†	**0.57 [0.41, 0.70]**	0.53 [0.41, 0.62]	0.48 [0.36, 0.58]	0.52 [0.43, 0.61]
MMLU (HELM)	**0.49 [0.36, 0.59]**	0.18 [0.04, 0.28]†	0.48 [0.27, 0.60]	0.45 [0.33, 0.55]	0.42 [0.30, 0.52]	0.44 [0.25, 0.56]

Table 11: Kendall τ_b, extrapolation, $n = 100$.

Benchmark	Random mean	Random + ridge	Anchor + IRT	AIPW	MASS (latent)	MASS-Ens
ARC-Challenge*	0.50 [0.42, 0.58]	0.37 [0.28, 0.46]†	**0.55 [0.48, 0.62]**	0.48 [0.41, 0.56]†	0.54 [0.48, 0.61]	0.55 [0.48, 0.61]
BBH	0.61 [0.56, 0.65]	0.39 [0.33, 0.44]†	0.60 [0.53, 0.65]	0.60 [0.55, 0.65]	0.61 [0.57, 0.66]	**0.61 [0.56, 0.65]**
GPQA*	**0.43 [0.38, 0.48]**†	0.30 [0.23, 0.35]	0.39 [0.33, 0.46]	0.43 [0.39, 0.47]†	0.41 [0.36, 0.46]	0.37 [0.33, 0.42]
IFEval*	**0.82 [0.79, 0.84]**	0.75 [0.72, 0.78]†	0.79 [0.76, 0.82]†	0.81 [0.78, 0.83]	0.79 [0.76, 0.81]†	0.81 [0.79, 0.83]
MMLU-Pro	0.59 [0.52, 0.64]†	0.44 [0.38, 0.49]†	0.60 [0.54, 0.64]†	0.58 [0.51, 0.63]†	0.57 [0.50, 0.63]†	**0.63 [0.57, 0.67]**
MuSR	0.33 [0.28, 0.40]†	0.39 [0.31, 0.46]	**0.42 [0.34, 0.49]**	0.34 [0.28, 0.39]	0.38 [0.31, 0.44]	0.38 [0.32, 0.45]
MATH L5 (OLL)	**0.71 [0.64, 0.75]**	0.58 [0.51, 0.64]†	0.69 [0.60, 0.75]	0.69 [0.63, 0.73]	0.68 [0.61, 0.72]	0.69 [0.64, 0.74]
CommonSense (HELM)	0.50 [0.37, 0.62]	0.36 [0.24, 0.48]†	0.46 [0.14, 0.67]	0.46 [0.34, 0.56]†	0.55 [0.42, 0.66]	**0.58 [0.44, 0.69]**
GSM8K (HELM)*	0.66 [0.54, 0.76]	0.54 [0.40, 0.67]†	**0.73 [0.62, 0.82]**†	0.63 [0.49, 0.73]	0.65 [0.50, 0.76]	0.65 [0.52, 0.76]
LegalBench (HELM)	0.49 [0.34, 0.59]	0.40 [0.29, 0.49]	0.45 [0.29, 0.56]	0.47 [0.31, 0.58]	**0.49 [0.29, 0.62]**	0.41 [0.25, 0.54]
MATH (HELM)	**0.71 [0.57, 0.79]**†	0.67 [0.54, 0.77]	0.69 [0.47, 0.84]	0.69 [0.56, 0.77]	0.64 [0.44, 0.78]	0.62 [0.44, 0.74]
MedQA (HELM)	0.64 [0.52, 0.74]	0.35 [0.23, 0.47]†	0.63 [0.44, 0.74]	0.63 [0.50, 0.73]	**0.64 [0.54, 0.73]**	0.63 [0.51, 0.70]
MMLU (HELM)	**0.66 [0.55, 0.77]**	0.31 [0.17, 0.43]†	0.65 [0.48, 0.77]	0.63 [0.51, 0.74]	0.62 [0.49, 0.74]	0.64 [0.52, 0.76]

Table 12: Kendall τ_b, interpolation, $n = 50$.

Benchmark	Random mean	Random + ridge	Anchor + IRT	AIPW	MASS (latent)	MASS-Ens
ARC-Challenge*	0.80 [0.78, 0.82]	**0.80 [0.78, 0.82]**	0.79 [0.77, 0.81]	0.79 [0.77, 0.80]	0.80 [0.78, 0.81]	0.80 [0.78, 0.81]
BBH	0.71 [0.68, 0.74]	**0.79 [0.77, 0.80]**†	0.72 [0.71, 0.74]	0.71 [0.69, 0.72]	0.71 [0.69, 0.72]	0.72 [0.70, 0.73]
GPQA*	0.37 [0.32, 0.43]	**0.58 [0.55, 0.61]**†	0.38 [0.35, 0.40]	0.35 [0.31, 0.40]	0.35 [0.33, 0.38]	0.38 [0.35, 0.41]
IFEval*	0.84 [0.83, 0.85]	**0.85 [0.84, 0.85]**	0.83 [0.83, 0.84]	0.84 [0.83, 0.84]	0.84 [0.83, 0.85]	0.84 [0.84, 0.85]
MMLU-Pro	0.78 [0.76, 0.80]	**0.81 [0.79, 0.82]**†	0.78 [0.77, 0.79]	0.77 [0.76, 0.78]	0.77 [0.76, 0.78]†	0.78 [0.77, 0.79]
MuSR	0.39 [0.33, 0.46]	**0.59 [0.58, 0.61]**†	0.44 [0.42, 0.48]	0.41 [0.38, 0.45]†	0.44 [0.41, 0.48]	0.46 [0.44, 0.48]
MATH L5 (OLL)	0.74 [0.72, 0.76]†	0.71 [0.69, 0.73]†	**0.77 [0.76, 0.79]**	0.72 [0.70, 0.74]†	0.76 [0.75, 0.78]	0.77 [0.75, 0.78]
CommonSense (HELM)	0.77 [0.71, 0.82]	0.76 [0.70, 0.81]†	0.77 [0.74, 0.80]†	0.76 [0.70, 0.81]†	0.80 [0.75, 0.83]	**0.81 [0.77, 0.84]**
GSM8K (HELM)*	**0.88 [0.86, 0.90]**	0.86 [0.84, 0.88]	0.87 [0.85, 0.90]	0.87 [0.85, 0.89]	0.87 [0.86, 0.89]	0.87 [0.85, 0.89]
LegalBench (HELM)	0.70 [0.67, 0.74]	**0.73 [0.69, 0.77]**	0.70 [0.66, 0.75]	0.70 [0.66, 0.75]	0.71 [0.67, 0.74]	0.72 [0.68, 0.76]
MATH (HELM)	**0.88 [0.85, 0.90]**	0.87 [0.84, 0.88]	0.85 [0.83, 0.87]†	0.86 [0.84, 0.88]	0.88 [0.85, 0.90]	0.88 [0.85, 0.90]
MedQA (HELM)	0.80 [0.78, 0.83]	0.80 [0.76, 0.82]	**0.84 [0.81, 0.86]**†	0.79 [0.75, 0.82]	0.81 [0.77, 0.84]	0.81 [0.78, 0.84]
MMLU (HELM)	0.76 [0.72, 0.81]	0.74 [0.69, 0.78]	0.76 [0.72, 0.80]	0.73 [0.68, 0.77]	**0.78 [0.75, 0.80]**	0.77 [0.73, 0.80]

Table 13: Kendall τ_b, interpolation, $n = 100$.

Benchmark	Random mean	Random + ridge	Anchor + IRT	AIPW	MASS (latent)	MASS-Ens
ARC-Challenge*	0.84 [0.83, 0.86]	0.85 [0.84, 0.87]	**0.86 [0.85, 0.87]**	0.84 [0.82, 0.85]†	0.86 [0.84, 0.87]	0.86 [0.85, 0.87]
BBH	0.78 [0.77, 0.80]	**0.85 [0.84, 0.86]**†	0.80 [0.79, 0.81]	0.78 [0.77, 0.79]†	0.78 [0.77, 0.79]†	0.80 [0.79, 0.80]
GPQA*	0.47 [0.42, 0.51]	**0.66 [0.63, 0.67]**†	0.49 [0.47, 0.51]	0.46 [0.42, 0.49]	0.48 [0.46, 0.50]	0.49 [0.46, 0.51]
IFEval*	0.89 [0.88, 0.90]	0.89 [0.88, 0.90]	0.88 [0.87, 0.89]†	0.89 [0.88, 0.89]	**0.89 [0.89, 0.90]**	0.89 [0.88, 0.90]
MMLU-Pro	0.82 [0.81, 0.83]†	**0.85 [0.84, 0.86]**†	0.83 [0.82, 0.83]†	0.82 [0.81, 0.83]†	0.84 [0.83, 0.85]	0.84 [0.83, 0.84]
MuSR	0.51 [0.45, 0.55]†	**0.68 [0.67, 0.70]**†	0.58 [0.55, 0.59]	0.56 [0.54, 0.57]†	0.58 [0.56, 0.60]	0.59 [0.57, 0.61]
MATH L5 (OLL)	0.80 [0.79, 0.82]	0.78 [0.76, 0.79]†	**0.84 [0.83, 0.85]**†	0.78 [0.76, 0.80]†	0.82 [0.81, 0.83]	0.82 [0.80, 0.83]
CommonSense (HELM)	0.82 [0.79, 0.86]†	0.82 [0.79, 0.86]†	0.85 [0.82, 0.87]†	0.81 [0.76, 0.85]†	0.86 [0.84, 0.88]	**0.88 [0.86, 0.90]**
GSM8K (HELM)*	**0.91 [0.90, 0.93]**	0.90 [0.88, 0.92]	0.89 [0.86, 0.92]	0.91 [0.90, 0.92]	0.90 [0.87, 0.92]	0.91 [0.90, 0.93]
LegalBench (HELM)	0.73 [0.68, 0.77]†	0.77 [0.74, 0.79]	0.77 [0.73, 0.80]†	0.75 [0.72, 0.79]†	**0.82 [0.80, 0.85]**	0.81 [0.78, 0.84]
MATH (HELM)	0.92 [0.90, 0.93]†	0.91 [0.90, 0.93]	0.91 [0.89, 0.92]	0.91 [0.89, 0.93]	**0.92 [0.91, 0.94]**†	0.89 [0.88, 0.91]
MedQA (HELM)	0.87 [0.85, 0.89]	0.87 [0.85, 0.89]	0.87 [0.85, 0.89]	0.87 [0.85, 0.88]†	0.88 [0.86, 0.90]	**0.89 [0.87, 0.91]**
MMLU (HELM)	0.84 [0.81, 0.87]	0.81 [0.77, 0.86]	0.83 [0.80, 0.86]	0.83 [0.79, 0.85]	**0.87 [0.85, 0.89]**	0.85 [0.82, 0.87]

Table 14: Latent rank, $n = 100$, RMSE ratio to the random mean (10 repetitions).

Protocol	Method	rank $k = 1$	$k = 2$ (default)	$k = 4$
Extrapolation	Random items + AIPW	0.86 [0.84, 0.89]	0.86 [0.83, 0.89]	0.86 [0.83, 0.89]
	MASS (latent)	0.82 [0.77, 0.88]	0.84 [0.79, 0.90]	0.80 [0.75, 0.85]
Interpolation	Random items + AIPW	0.84 [0.81, 0.86]	0.83 [0.80, 0.86]	0.82 [0.79, 0.85]
	MASS (latent)	0.78 [0.75, 0.82]	0.77 [0.74, 0.80]	0.78 [0.74, 0.82]

Table 15: Extrapolation, $n = 100$ (as Table 1).

Method	RMSE ratio, all 13	RMSE ratio, 9 non-dev.	wins (of 13)	Kendall τ_b
Sources only	5.63 [3.92, 7.99]	5.47 [3.79, 7.76]	0	– (constant)
Random mean	1.00 [1.00, 1.00]	1.00 [1.00, 1.00]	–	**0.59 [0.52, 0.66]**
Stratified	0.87 [0.84, 0.91]	0.86 [0.83, 0.91]	13	**0.58 [0.50, 0.65]**
Calibrated random	1.09 [1.00, 1.18]	1.08 [0.98, 1.18]	3	**0.59 [0.52, 0.66]**
Random + ridge	1.65 [1.38, 1.96]	1.65 [1.31, 2.04]	2	0.45 [0.38, 0.53]
Anchor points	1.26 [0.93, 1.70]	1.16 [0.82, 1.65]	5	0.50 [0.41, 0.57]
IRT (random)	0.85 [0.82, 0.88]	0.84 [0.80, 0.87]	13	0.57 [0.51, 0.64]
IRT (D-opt.)	1.00 [0.77, 1.26]	**0.93 [0.68, 1.22]**	7	**0.51 [0.40, 0.61]**
Anchor + IRT	**0.80 [0.75, 0.84]**	**0.78 [0.72, 0.84]**	13	**0.59 [0.53, 0.66]**
AIPW	0.86 [0.83, 0.89]	0.85 [0.81, 0.88]	13	0.57 [0.50, 0.64]
AIPW + ridge (cross-fit)	0.94 [0.90, 0.98]	0.92 [0.87, 0.98]	11	0.54 [0.47, 0.61]
PPI	0.86 [0.83, 0.89]	0.85 [0.81, 0.88]	13	0.57 [0.50, 0.64]
AIPW-kNN	0.87 [0.82, 0.93]	0.86 [0.80, 0.92]	11	0.58 [0.51, 0.64]
AIPW-Ens	0.84 [0.81, 0.87]	0.83 [0.79, 0.87]	13	0.58 [0.51, 0.64]
MASS (latent)	0.84 [0.79, 0.90]	0.84 [0.78, 0.91]	11	**0.58 [0.52, 0.64]**
MASS-Ens	**0.78 [0.74, 0.83]**	**0.76 [0.70, 0.82]**	13	**0.58 [0.52, 0.65]**

Table 16: Interpolation, $n = 100$.

Method	RMSE ratio, all 13	RMSE ratio, 9 non-dev.	wins (of 13)	Kendall τ_b
Sources only	3.13 [2.17, 4.36]	3.11 [2.13, 4.38]	2	– (constant)
Random mean	1.00 [1.00, 1.00]	1.00 [1.00, 1.00]	–	**0.79 [0.71, 0.85]**
Stratified	0.84 [0.82, 0.87]	0.84 [0.82, 0.86]	13	**0.79 [0.71, 0.85]**
Calibrated random	0.80 [0.75, 0.85]	0.82 [0.77, 0.86]	13	**0.79 [0.71, 0.85]**
Random + ridge	**0.72 [0.62, 0.83]**	**0.75 [0.63, 0.87]**	12	**0.82 [0.78, 0.86]**
Anchor points	1.08 [0.93, 1.28]	1.04 [0.85, 1.28]	6	**0.79 [0.68, 0.87]**
IRT (random)	0.82 [0.80, 0.85]	0.82 [0.79, 0.86]	13	0.78 [0.71, 0.85]
IRT (D-opt.)	1.27 [1.08, 1.53]	1.34 [1.10, 1.70]	3	**0.76 [0.66, 0.84]**
Anchor + IRT	**0.81 [0.76, 0.86]**	0.80 [0.75, 0.86]	13	**0.80 [0.73, 0.86]**
AIPW	0.83 [0.80, 0.86]	0.82 [0.79, 0.86]	13	0.78 [0.71, 0.85]
AIPW + ridge (cross-fit)	0.86 [0.82, 0.90]	0.85 [0.80, 0.91]	13	0.78 [0.70, 0.84]
PPI	0.83 [0.80, 0.85]	0.82 [0.79, 0.86]	13	0.79 [0.71, 0.85]
AIPW-kNN	**0.79 [0.76, 0.83]**	0.79 [0.74, 0.84]	13	0.79 [0.72, 0.85]
AIPW-Ens	**0.80 [0.77, 0.83]**	0.79 [0.76, 0.83]	13	0.79 [0.72, 0.85]
MASS (latent)	**0.77 [0.74, 0.80]**	**0.76 [0.73, 0.79]**	13	**0.81 [0.74, 0.87]**
MASS-Ens	**0.75 [0.71, 0.79]**	**0.73 [0.69, 0.79]**	13	**0.81 [0.74, 0.87]**

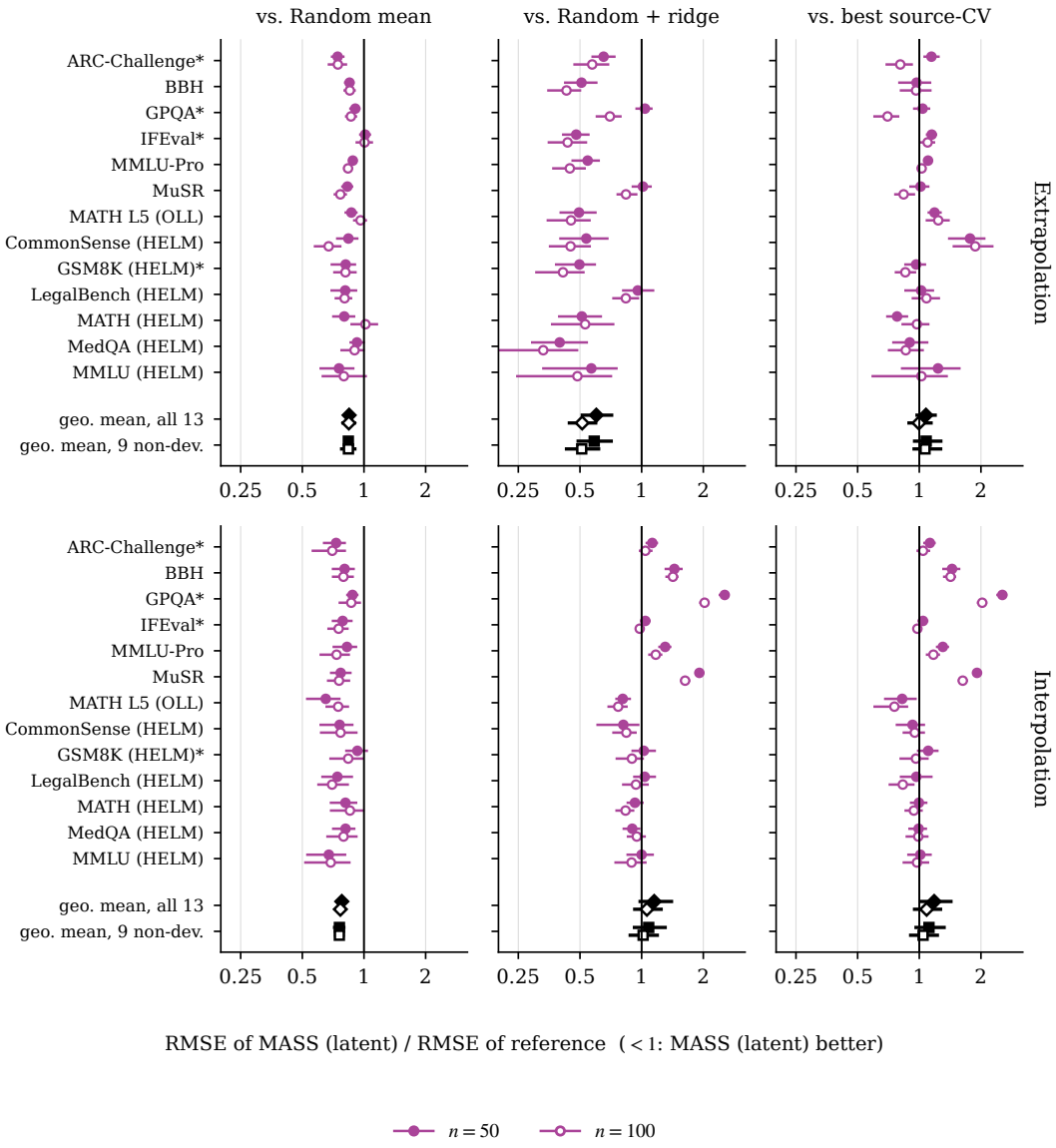


Figure 4: As Figure 2 for the latent-only MASS.

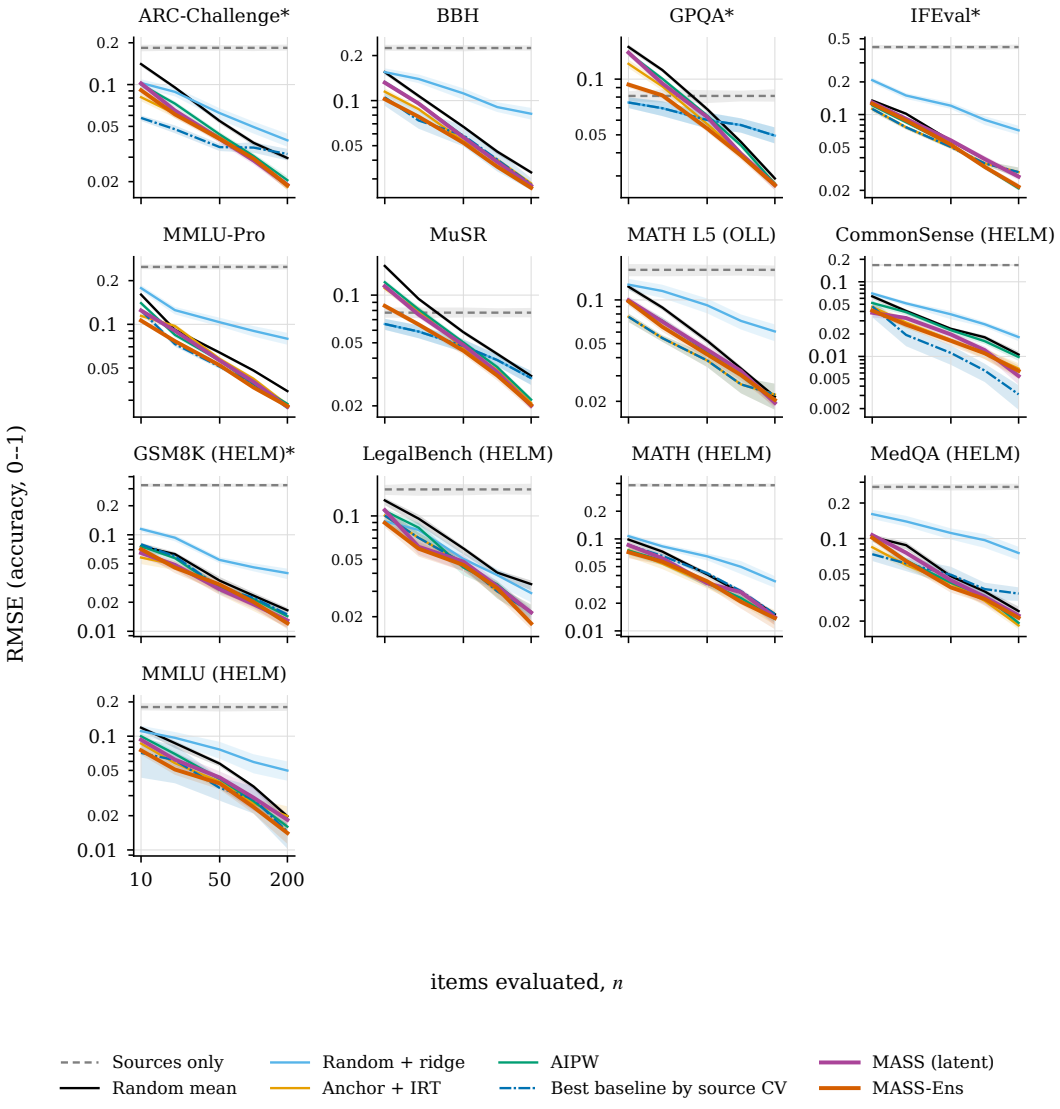


Figure 5: Per-benchmark RMSE versus n, extrapolation.

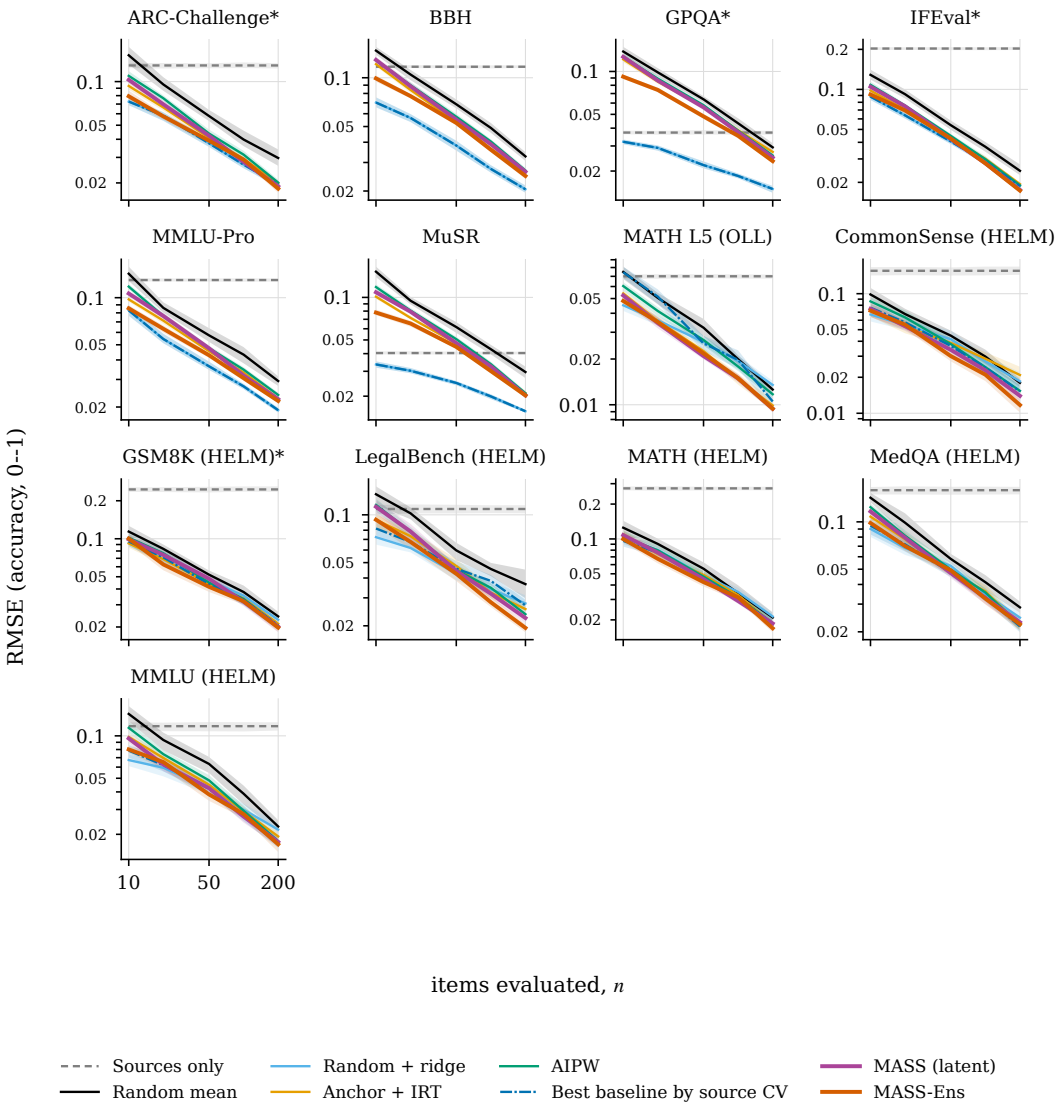


Figure 6: Per-benchmark RMSE versus n, interpolation.

D Additional figures

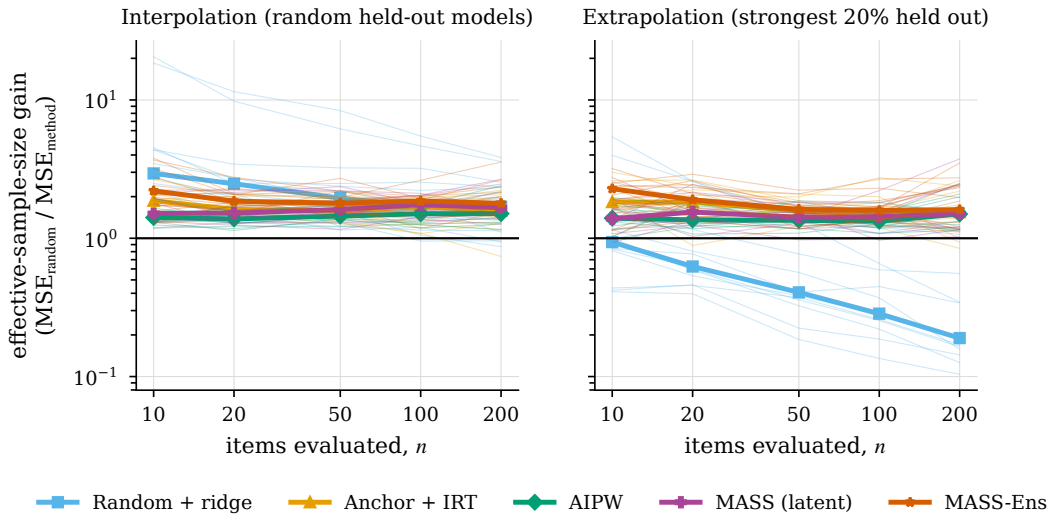


Figure 7: Effective-sample-size gain, $(\mathrm{RMSE}_{\mathrm{random}}/\mathrm{RMSE}_{\mathrm{method}})^2$, per benchmark (thin) and median over benchmarks (thick). Random + ridge falls below 1 under extrapolation.

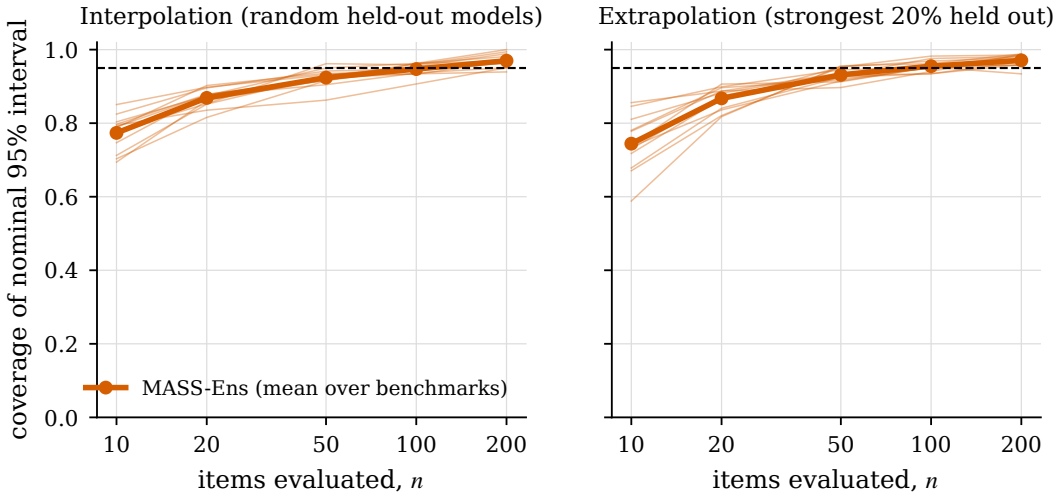


Figure 8: Share of target models whose true score lies within ± 1.96 reported standard errors of MASS-Ens; thin lines are benchmarks, thick line the mean over benchmarks; dashed = nominal 95%.

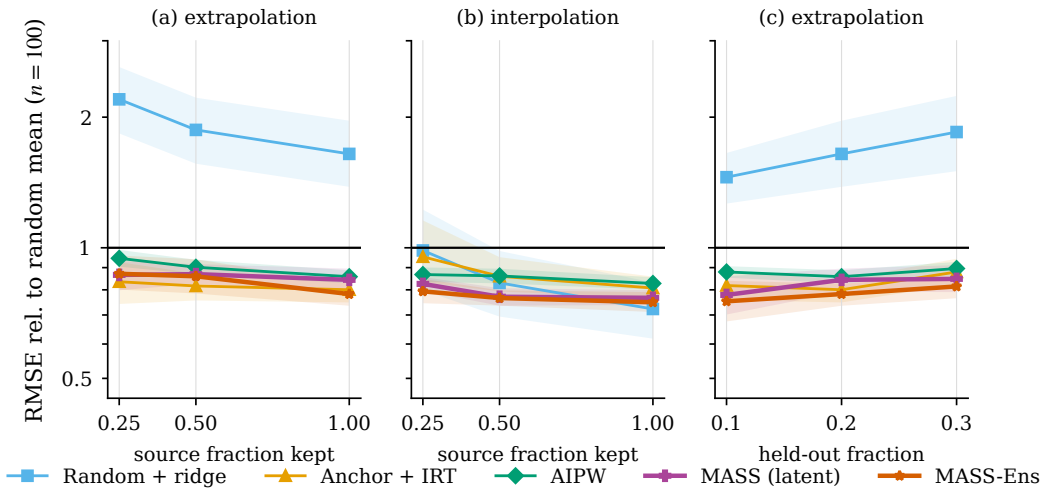


Figure 9: RMSE ratio to the random mean ($n = 100$) versus the fraction of source models kept (a, b) and the held-out fraction of strongest models under extrapolation (c).

E Reproducibility

Environment, data version (commit `55faff49`), seeds, commands, expected run times and the map from each table and figure to its generating script are in `README.md` and `code/README.md` of the supplementary code. The paper contains no number that is not produced by those scripts from `results/`. Statements in the text that quote a number (ratios, coverage, bias, effective-sample-size gains) are read from `results/cells.csv`, `aggregate.csv`, `coverage.csv`, `ess_gain.csv` and `paired.csv`. Contact with the forbidden-content rule of the task is recorded in the reproduction notes.

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